

1) Merge Sort vs Radix Sort

Basis	Merge Sort	Radix Sort
a) Type of Sorting	Comparison based Sorting	Non-comparison-based Sorting
b) Input limitations	Can sort almost any data type that has a comparison operation	mainly suitable for integers, strings, or Fixed-format key
c) Performance on large datasets	$O(n \log n)$, reliable for large Datasets	$O(d \times (n+k))$, can be faster when the number of digits d is small

Conclusion: Merge sort is more general purpose, while Radix sort can be faster for suitable integer/string data

2) Methods of Building a heap

a) Bottom-up heap construction

- *> Start with all elements in an array
- *> Begin from the last non-leaf node.
- *> Apply heapify repeatedly toward the root.

*> The complexity: $O(n)$

b) Repeated insertion into heap

- *> start with an empty heap
- *> Insert each element one by one
- *> After every insertion, restore the heap property using up-heap/bubble-up
- *> Time complexity: $O(n)$

Which is more efficient?

Bottom-up heap construction is more efficient.

* > Bottom-up method: $O(n)$

* > Repeated insertion: $O(n \log n)$

$\therefore$ For Building a heap from a complete set of elements, bottom-up construction is preferred because it takes linear time.

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